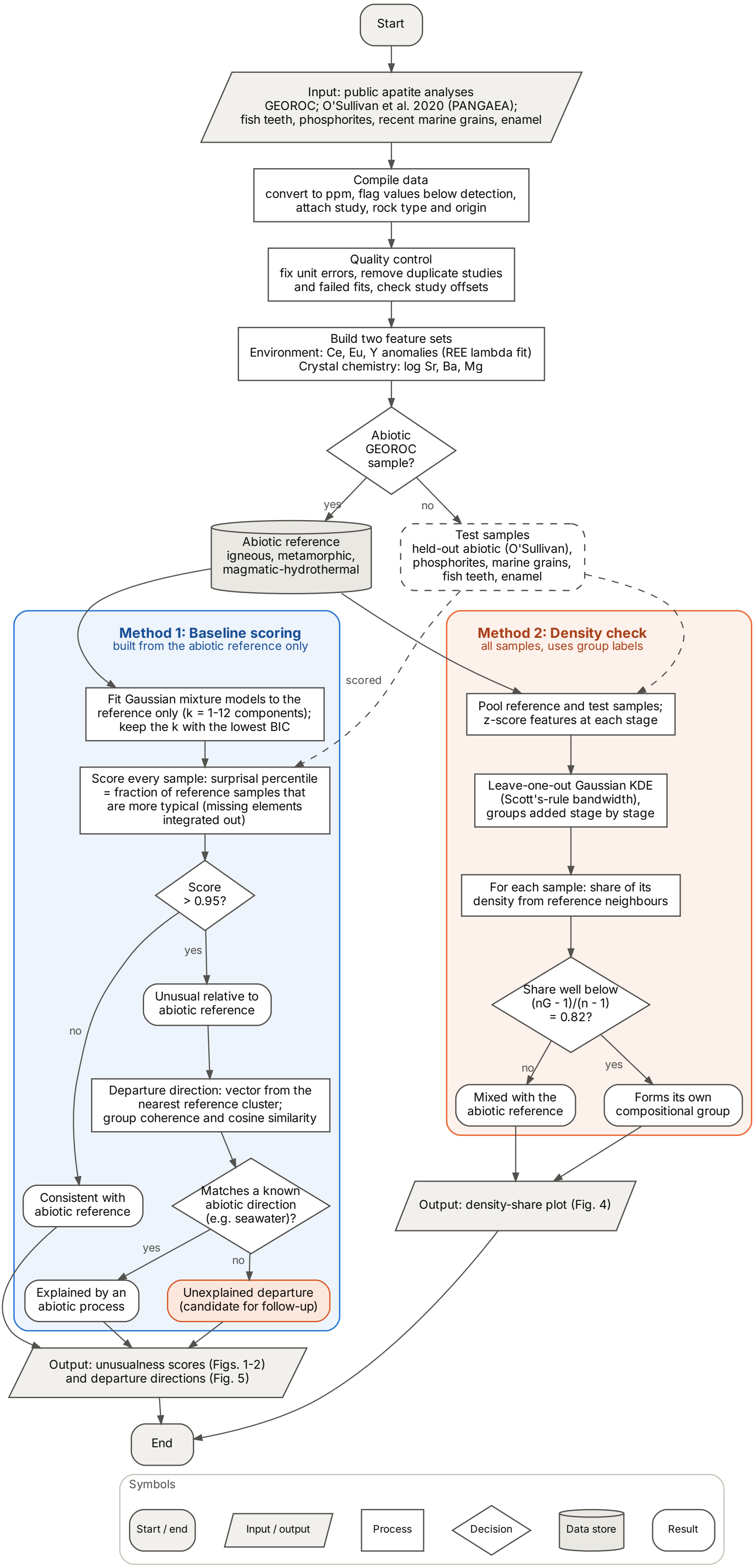
Start
Input: public apatite analyses
GEOROC; O'Sullivan et al. 2020 (PANGAEA);
fish teeth, phosphorites, recent marine grains, enamel
Compile data
convert to ppm, flag values below detection,
attach study, rock type and origin
Quality control
fix unit errors, remove duplicate studies
and failed fits, check study offsets
Build two feature sets
Environment: Ce, Eu, Y anomalies (REE lambda fit)
Crystal chemistry: log Sr, Ba, Mg
Abiotic GEOROC sample?
yes
no
Abiotic reference
igneous, metamorphic,
magmatic-hydrothermal
Test samples
held-out abiotic (O'Sullivan),
phosphorites, marine grains,
fish teeth, enamel
Method 1: Baseline scoring
built from the abiotic reference only
scored
Fit Gaussian mixture models to the
reference only (k = 1-12 components);
keep the k with the lowest BIC
Score every sample: surprisal percentile
= fraction of reference samples that
are more typical (missing elements
integrated out)
Score
> 0.95?
yes
no
Unusual relative to
abiotic reference
Departure direction: vector from the
nearest reference cluster;
group coherence and cosine similarity
Consistent with
abiotic reference
Matches a known
abiotic direction
(e.g. seawater)?
yes
no
Explained by an
abiotic process
Unexplained departure
(candidate for follow-up)
Method 2: Density check
all samples, uses group labels
Pool reference and test samples;
z-score features at each stage
Leave-one-out Gaussian KDE
(Scott's-rule bandwidth),
groups added stage by stage
For each sample: share of its
density from reference neighbours
Share well below
(nG - 1)/(n - 1)
= 0.82?
no
yes
Mixed with the
abiotic reference
Forms its own
compositional group
Output: density-share plot (Fig. 4)
Output: unusualness scores (Figs. 1-2)
and departure directions (Fig. 5)
End
Symbols
Start / end
Input / output
Process
Decision
Data store
Result